

CHAPTER 35

Group Level Analyses

Approach 1

Exploratory

→ kind of situation where we don't have a real hypothesis / maybe we have a bit of expectation on which conditions we should be comparing / which electrodes / brain regions, but we don't have a hard constraint for which frequencies / which timepoints we want to constrain ourselves to

Approach 2a/b

Hypothesis-driven

→ case where we have a very strong theory / a priori motivation based on some prior data or some theoretic framework where we have hard constraint that tells us which frequencies are most relevant and which time windows are most relevant

Approach 1:

Idea: permutation testing and cluster correction, just like with mass-univariate approach
↳ testing lots and lots of individual pixels

Advantages: Don't worry about assumptions.
lower chance of bias / subjectivity
& open to exploratory discovery

Disadvantages: Designed for 2-test or correlation comparisons.
potential lower sensitivity to small but theoretically motivated effects

Approach 2a:

Average all of conditions together
(all of experiment conditions, all of subjects everything together) and then look at Time Frequency power plot of

grand grand total average relative to baseline and pick some time freq window & then this window gets applied to each individual subject & each individual condition and then average together the power values / phase clustering / connectivity (whatever is the feature of data that we are working on) in that time freq window to create a single number to build table like below

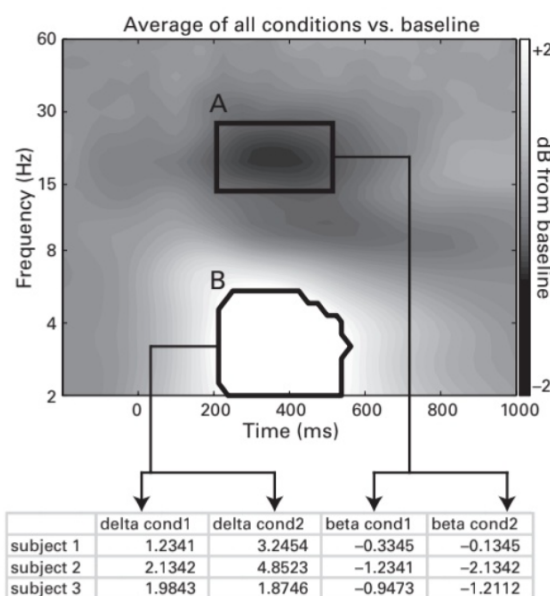


Figure 35.1

Illustration of extracting data for group analysis strategy 2a. Based on either visual inspection (box A) or a statistical contrast orthogonal to the hypotheses (area B), time-frequency regions are defined. In practice, either but not both of these methods should be used within a study; both are shown here for illustration. The activity from all pixels in each region is averaged separately for each subject and each condition and then placed into a matrix that can be exported to a statistics program. Although this figure illustrates only two conditions (cond1 and cond2), this approach is well suited for factorial designs.

Idea: Extract average TF data from one/more windows

Advantages: suitable for factorial designs - Hypothesis driven.
Minimal multiple Comparisons issues (Bonferroni OK)

Disadvantages: potential for bias.
can miss important but unpredicted results

→ Maximal sensitivity to potentially small effects

Approach 2b:

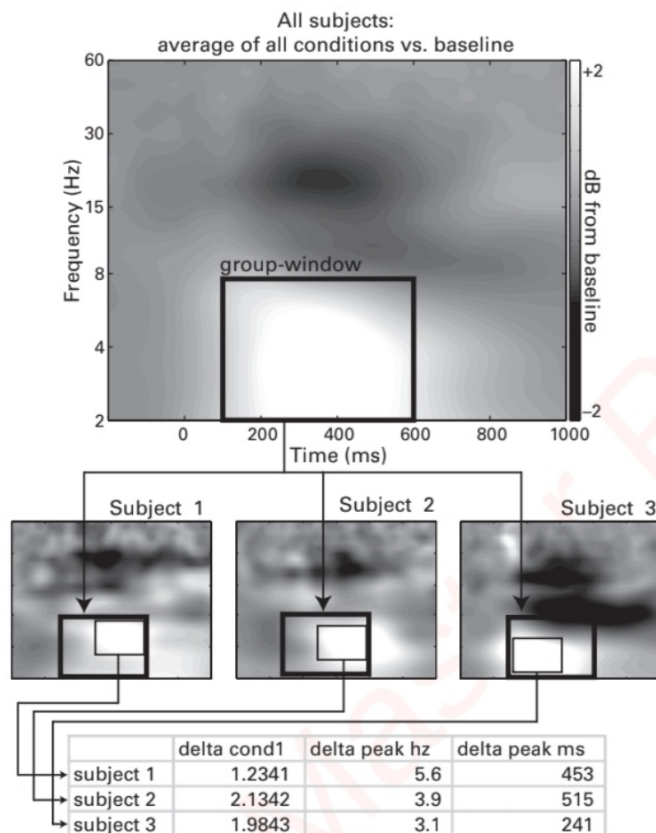


Figure 35.3

Illustration of group-level analysis strategy 2b. A relatively large window is drawn based on the average of all conditions and all subjects (the group window). From that window a smaller window within the group window is selected for each subject individually, based on the subject-specific time-frequency peak. This method allows you to extract not only the average activity for each condition and subject but also peak time and frequency points.

we draw a larger window, which is still the average over all experiment conditions and the average over all subjects, then within this large window, we pick smaller windows that can be applied to each individual subject

we can also calculate peak freq and peak time with approach 2b, which is not possible with approach 2a

Idea % Extract average TF data from one/more windows, optimized for each subject

Advantages % suitable to factorial designs • Hypothesis driven. Minimal multiple comparisons issues (Bonferroni OK). can also extract peak time/frequency per subject

Disadvantages % potential for bias. can be driven by noise at single subject level can miss important but unpredicted results

Circular inference (double-dipping):

circular inference: selecting data in a way that biases the test statistic (and thus, interpretation) towards a particular result.

Example:

- Steps:
- 1) Generate 20 "normal" random numbers
 - 2) select the 10 largest numbers
 - 3) Run a t-test against 0
 - 4) Repeat 100 times

Results: ~90% of t values were significant at $p < 0.05$

→ selected samples in a way that biased test

Avoiding circular inference:

- 1) Test everything, correct appropriately
- 2) select data, test orthogonally